CSA 0921 - PROGRAMMING IN JAVA

CO1 - AT3 - PSEUDO CODE WRITING

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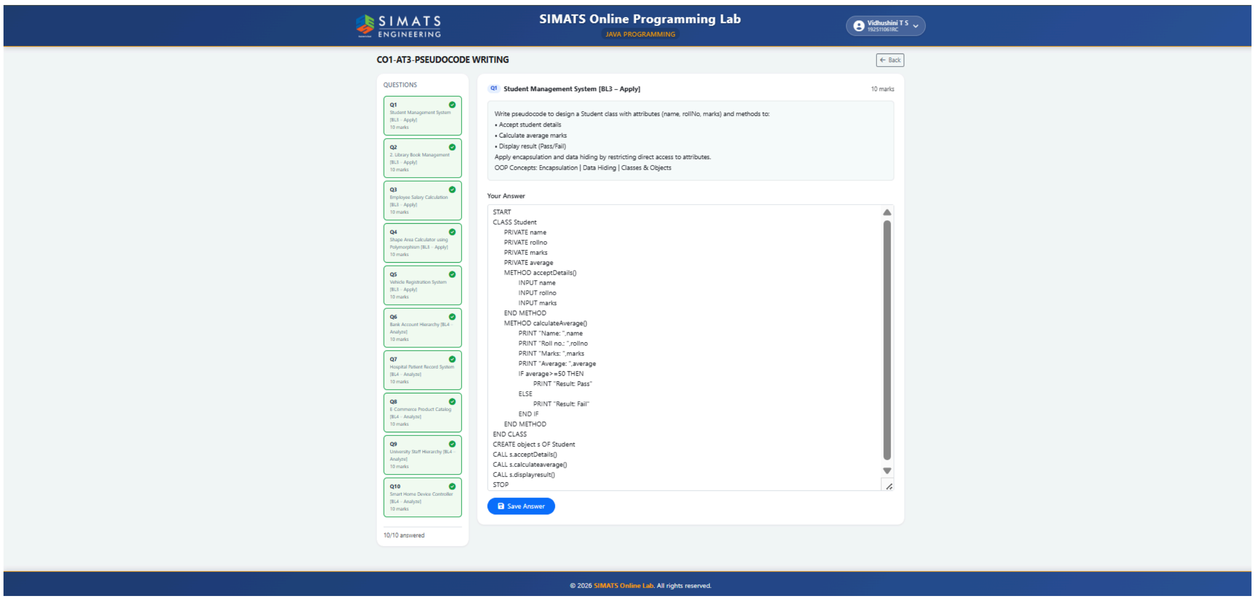

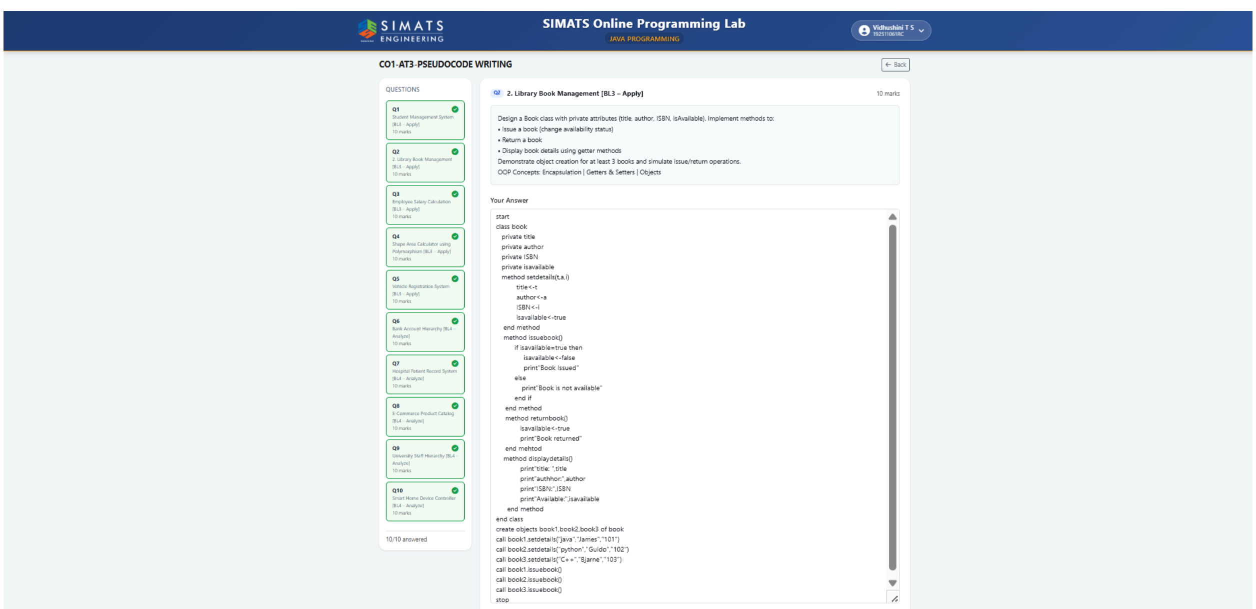

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QUESTIONS

- Q1 Student Management System [BL3 – Apply] 10 marks
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- Q10 Smart Home Device Controller [BL4 – Analyze] 10 marks

10/10 answered

Q3 **Employee Salary Calculation [BL3 – Apply]** 10 marks

Write pseudocode for an Employee class with encapsulated attributes (empId, name, basicPay, designation). Implement methods to: •
Calculate gross salary (basic + HRA + DA)
• Calculate net salary after deductions (PF, tax)
• Display pay slip
Apply data hiding so salary components are not directly accessible.
OOP Concepts: Encapsulation | Data Hiding | Methods

Your Answer

```
START
CLASS Employee
    PRIVATE empId, name, designation, basicPay
    PRIVATE HRA, DA, PF, tax, grossSalary, netSalary
    METHOD acceptDetails()
        INPUT empId, name, designation, basicPay
    END METHOD
    METHOD calculateGross()
        HRA<- basicPay*0.20
        DA<- grossSalary*0.10
        grossSalary<-basicPay+HRA+DA
    END METHOD
    METHOD calculateNet()
        PF<-basicPay*0.12
        tax<-grossSalary*0.5
        netSalary<-grossSalary-PF-tax
   END METHOD
   METHOD displayPaySlip()
       PRINT empId,name,desgination
       PRINT grossSalary
       PRINT netSalary
   END METHOD
END CLASS
CREATE object e
CALL e.acceptDetails()
CALL e.calculateGross()
CALL e.calculateNet()
CALL e.displayPaySlip()
STOP
```

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10/10 answered

Q4 **Shape Area Calculator using Polymorphism [BL3 – Apply]** 10 marks

Create a Shape base class with an overridable calculateArea() method. Apply this to:
• Circle — area using radius
• Rectangle — area using length and breadth
• Triangle — area using base and height
Use method overriding to demonstrate runtime polymorphism. Call calculateArea() for each object.
OOP Concepts: Inheritance | Polymorphism | Method Overriding

Your Answer

```
START
CLASS Shape
    METHOD calculateArea()
        PRINT "Area"
    END METHOD
END CLASS
CLASS Circle EXTENDS Shape
    radius
    OVERRIDE calculateArea()
        area<-3.14*radius*radius
        PRINT area
    END METHOD
END CLASS
CLASS Rectangle EXTENDS Shape
    length, breadth
    OVERRIDE calculateArea()
        area<- length*breadth
        PRINT area
    END METHOD
END CLASS
CLASS Triangle EXTENDS Shape
    base, height
    OVERRIDE calculateArea()
       area<- 0.5*base*height
       PRINT area
    END METHOD
END CLASS
CREATE Circle, Rectangle, Triangle objects
CALL calculateArea() for each object
STOP
```

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10/10 answered

Q5 Vehicle Registration System [BL3 – Apply] 10 marks

Design a Vehicle class with attributes (regNo, owner, fuelType). Extend it into:
• TwoWheeler — with engine capacity
• FourWheeler — with seating capacity and AC status
Apply inheritance to reuse common attributes and override a displayDetails() method in each subclass.
OOP Concepts: Inheritance | Method Overriding | Subclass

Your Answer

```
START
CLASS Vehicle
    regno, owner, fuelType
    METHOD displayDetails()
        PRINT regno, owner, fuelType
    END METHOD
END CLASS
CLASS TwoWheeler EXTENDS Vehicle
    engineCapacity
     OVERRIDE displayDetails()
          PRINT regno, owner, fuelType, engineCapacity
     END METHOD
END CLASS
CLASS FourWheeler EXTENDS Vehicle
    seatingCapacity, ACstatus
    OVERRIDE displayDetails()
         PRINT regno, owner, fuelType
         PRINT seatingCapacity, ACstatus
     END METHOD
END CLASS
CREATE TwoWheeler object
CREATE FourWheeler object
CALL displayDetails()
STOP
```

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10/10 answered

Q6 Bank Account Hierarchy [BL4 – Analyze] 10 marks

Develop pseudocode for a BankAccount superclass and two subclasses — SavingsAccount (with interest calculation) and CurrentAccount (with overdraft facility). Analyze and implement:
• Inheritance of common attributes (accNo, balance, holder)
• Method overriding for withdrawal behavior
• How overdraft protection differs from savings limit
Compare the behavior of withdraw() across both subclasses and justify the design choices.
OOP Concepts: Inheritance | Method Overriding | Polymorphism

Your Answer

```
START
CLASS BankAccount
     accno, holder, balance
     METHOD withdraw(amount)
         balance<- balance-amount
     END METHOD
END CLASS
CLASS SavingsAccount EXTENDS BankAccount
    OVERRIDE withdraw(amount)
         IF amount<=Balance THEN
            balance<-balance-amount
         ELSE
            PRINT "Insufficient Balance"
         END IF
    END METHOD
END CLASS
CLASS CurrentAccount EXTENDS BankAccount
    overdraftLimit
     OVERRIDE withdraw(amount)
         IF amount<= balance+overdraftLimit THEN
             balance<- balance-amount
         ELSE
             PRINT "Overdraft Limit Exceeded"
         END IF
     END METHOD
END CLASS
CREATE SavingsAccount object
CREATE CurrentAccount object
CALL withdraw()
STOP
```

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10/10 answered

Q7 Hospital Patient Record System [BL4 – Analyze]

10 marks

Design a Patient base class and analyze how it extends into:
• InPatient — with ward number, admission date, and daily charges
• OutPatient — with appointment date and consultation fee
Analyze encapsulation requirements for sensitive data (diagnosis, prescription). Override a generateBill() method for each type and compare the billing logic differences. OOP Concepts: Encapsulation | Inheritance | Polymorphism

Your Answer

```
START
CLASS Patient
    PRIVATE patientId, name, diagnosis, prescription
    METHOD generateBill()
        PRINT "Patient Bill"
    END METHOD
END CLASS
CLASS InPatient EXTENDS Patient
     wardNo, admissionDate, dailyCharges
      OVERRIDE generateBill()
          bill<-dailyCharges*days
          PRINT bill
      END METHOD
END CLASS
CLASS OutPatient EXTENDS Patient
     appointmentDate, consultantionFee
     OVERRIDE generateBill()
         PRINT consultationFee
      END METHOD
END CLASS
CREATE InPatient object
CREATE OutPatient object
CALL generateBill()
STOP
```

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10/10 answered

Q8 E-Commerce Product Catalog [BL4 – Analyze]

10 marks

Analyze and design a Product class hierarchy for an e-commerce system:
• Electronics — with warranty, brand, voltage
• Clothing — with size, fabric, gender
• Grocery — with expiryDate and weight
Override an applyDiscount() method for each category with different discount rules. Analyze why polymorphism makes the cart checkout process flexible and extensible. OOP Concepts: Polymorphism | Inheritance | Method Overriding

Your Answer

```
START
CLASS Product
    productId, name, price
    METHOD applyDiscount()
       PRINT price
    END METHOD
END CLASS
CLASS Electronics EXTENDS Product
    OVERRIDE applyDiscount()
        price<- price-(price*10/100)
        PRINT price
    END METHOD
END CLASS
CLASS Clothing EXTENDS Product
    OVERRIDE applyDiscount()
        price<- price-(price*20/100)
        PRINT price
     END METHOD
END CLASS
CLASS Grocery EXTENDS Product
     OVERRIDE applyDiscount()
         price<- price-(price*5/100)
         PRINT price
      END METHOD
END CLASS
CREATE Electronics, Clothing, Grocery objects
CALL applyDiscount()
STOP
```

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Q9 University Staff Hierarchy [BL4 – Analyze]

10 marks

Analyze the design of a Staff base class extended by TeachingStaff and NonTeachingStaff. Further extend TeachingStaff into Professor and Lecturer (multilevel inheritance). For each level:

- Identify which attributes should be private vs protected
- Override calculateAllowance() at each level
- Analyze how protected access enables inheritance without full exposure Justify the use of multilevel inheritance and explain how access modifiers enforce data hiding.

OOP Concepts: Multilevel Inheritance | Access Modifiers | Data Hiding

Your Answer

```
START
CLASS Staff
   PRIVATE staffId, name
   PROTECTED salary
   METHOD calculateAllowance()
       PRINT "Allowance"
    END METHOD
END CLASS
CLASS TeachingStaff EXTENDS Staff
     OVERRIDE calculateAllowance()
          PRINT salary*0.20
     END METHOD
END CLASS
CLASS Professor EXTENDS TeachingStaff
    OVERRIDE calculateAllowance()
        PRINT salary*0.30
     END METHOD
END CLASS
CLASS Lecturer EXTENDS TeachingStaff
     OVERRIDE calculateAllowance()
         PRINT salary*0.25
     END METHOD
END CLASS
CLASS NonTeachingStaff EXTENDS Staff
      OVERRIDE calculateAllowance()
          PRINT salary*0.15
       END METHOD
END CLASS
CREATE Professor, Lecturer, NonTeachingStaff objects
CALL calculateAllowance()
STOP
```

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10/10 answered

Q10 Smart Home Device Controller [BL4 – Analyze]

10 marks

Analyze and design a SmartDevice superclass with subclasses: SmartLight,
SmartThermostat, and SmartLock. Override an executeCommand(cmd) method in each, where the same command (e.g., 'on', 'off', 'status') produces device-specific behavior. Analyze:

- How polymorphism allows a single controller loop to manage all devices
- Security implications of data hiding in SmartLock (PIN must never be exposed) Design the class hierarchy and explain the OOP principles applied at each level.

OOP Concepts: Polymorphism | Encapsulation | Inheritance

Your Answer

```
START
CLASS SmartDevice
     METHOD executeCommand(cmd)
         PRINT cmd
     END METHOD
END CLASS
CLASS SmartLight EXTENDS SmartDevice
    OVERRIDE executeCommand(cmd)
         PRINT "Light"+cmd
     END METHOD
END CLASS
CLASS SmartThermostat EXTENDS SmartDevice
    OVERRIDE executeCommand(cmd)
          PRINT "Thermostat"+cmd
     END METHOD
END CLASS
CLASS SmartLock EXTENDS SamrtDevice
    PRIVATE PIN
    OVERRIDE executeCommand(cmd)
        PRINT"Lock"+cmd
    END METHOD
END CLASS
CREATE SmartLight object
CREATE SmartThermostat object
CREATE SmartLock object
CALL executeCommand("ON")
CALL executeCommand("OFF")
CALL executeCommand("STATUS")
STOP
```

Save Answer

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